

Models guide: choosing, storing and upgrading the AI models

The research assistant uses two AI models. This guide explains what they are, what you need to pay for to run them, and how to pick replacements on Hugging Face when better ones come out, even if the names there mean nothing to you yet.

You don't need an account on Hugging Face. The setup notebook downloads the models, and the cloud notebook installs everything else, each time it runs.

1. What you need to pay for

This project won't run on free accounts. You need two paid plans.

Google One (storage)

A free Google account has 15 GB of storage, shared between Gmail, Google Photos and Google Drive. The project needs more than that:

- **The AI models:** about 15 GB, stored once on your Drive.
- **Your literature:** every paper the forager finds is kept on Drive. A PDF is typically 1–5 MB, so 1,000 papers is roughly 2–5 GB, and the collection keeps growing.
- **The agent's notes:** summaries, logs and memories. Small, but they add up.

A Google One plan with **100 GB or more** is the practical minimum.

Google Colab (computing power)

The models run on Google's computers, through Colab. A paid Colab plan gives you:

- **A GPU with enough memory to hold the model** (see section 2), available when you need it. On the free plan, GPUs are not guaranteed and sessions are cut short.
- **High-RAM**, the setting the notebooks ask for, which is only available on paid plans.
- **Longer sessions**, so the agent can work through a large pile of papers in one go.

Plan names and prices change, so check the current options on the Colab website.

2. Three kinds of memory, and why they matter

Computers have three different places to keep things.

SSD (Solid-State Drive). The slowest of the three. On your laptop the SSD is permanent; on the Colab machine it's temporary and wiped when the session ends. At the start of each session, the agent copies

your papers from Google Drive onto Colab's SSD, and Ollama makes its own copy of the models there, so they can be reached quickly. Results are saved back to Google Drive, because Colab's SSD won't keep them.

RAM (Random Access Memory). The computer's working memory: much faster than the SSD, but it only holds what is in use right now, and it's emptied when the machine stops. It holds the notebook's program, the paper being read, and the model while it's being loaded onto the GPU.

GPU memory. The GPU (Graphics Processing Unit) is a separate chip that does the heavy maths of an AI model very fast, but only on things held in its *own* memory. A model has to be loaded into GPU memory before it can run. If the model doesn't fit, the part that doesn't fit stays in normal RAM and is run by the ordinary processor instead. It still works, but several times more slowly. **GPU memory should have spare room.** While it's working, the model also needs space for the text it's currently reading and some working space. Allow **2–3 GB on top of the model's file size.**

Google Drive. is separate from all three. It's permanent storage on Google's servers, shared by your laptop and Colab, and it's where the SSD copies come from.

3. The two models

	The Sorter (embeddings model)	The Writer (Large Language Model)
Currently	local-embedder.gguf (~0.4 GB file)	local-analyst.gguf (~14.3 GB file)
What it does	Decides where to cut each paper into chunks	Reads each chunk and writes the summary sentence; brainstorms search queries; condenses summaries into themes; updates the research focus

The writer: a language model

A language model, often called a "large language model" or "LLM", has learned from enormous amounts of text to read and write. It's the part of the pipeline that does the thinking. Bigger models generally write and reason better, but need more GPU memory.

The sorter: an embeddings model

An embeddings model doesn't write anything. It reads a sentence and gives back a set of numbers describing what the sentence is **about**, so, sentences about similar things are numerically close together.

The pipeline uses this to split papers sensibly, so that each piece then covers one idea, meaning cleaner material to summarise. Replacing the embeddings model never affects work already done.

4. Hugging Face in one short paragraph

[Hugging Face](#) is the world's main public library of AI models, for anyone to download. The setup notebook fetches the models from there, and you'll browse it when choosing new ones. You don't need an account for the models this guide recommends.

5. Choosing a model on Hugging Face

Hugging Face lists thousands of models with names like `Qwen3-14B-Instruct-GGUF` or `Llama-3.3-70B-Instruct-Q4_K_M.gguf`. They look like gibberish, but they follow a pattern.

Reading a model name

Take the current writer, `Mistral-Small-24B-Instruct-2501-GGUF`:

Part	Meaning	What to look for
<code>Mistral-Small</code>	The family (who made it) and version	Any well-known family is fine
<code>24B</code>	Size : 24 billion internal settings. Bigger = smarter but heavier	Decides whether it fits (see below)
<code>Instruct</code>	Trained to follow instructions . Also written <code>Chat</code> or <code>it</code>	Required . Avoid names with <code>Base</code>
<code>2501</code>	Release date (2025, January)	Newer is usually better
<code>GGUF</code>	The file format Ollama can run	Required

Words that mean "not for this project": `Base` (doesn't follow instructions), `Coder` (specialised for programming), `Vision` or `VL` (for images; works, but wastes space).

Reading the files inside

Open a model and click **Files and versions**. A GGUF model offers the same model in several sizes, from "compressed a lot" to "barely compressed". The letters in each file name tell you how much:

In the file name	Compression	Choose it when
<code>Q8_0</code>	Light (biggest file, best quality)	You have lots of GPU memory to spare
<code>Q6_K</code> , <code>Q5_K_M</code>	Moderate	You have some room to spare
<code>Q4_K_M</code>	Standard (the usual choice)	Almost always: good quality at a sensible size
<code>Q3_...</code> , <code>IQ3_...</code> , <code>Q2_...</code>	Heavy (noticeably worse)	Only if nothing else fits

A bigger model in `Q4_K_M` usually beats a smaller model in `Q8_0` of the same file size.

Will it fit?

Look at the file size shown next to the `Q4_K_M` file, and compare it with your GPU:

Colab GPU	GPU memory	Largest comfortable file size
T4	~15 GB	about 12 GB (the current 14.3 GB model is right at the limit)
L4	~22 GB	about 19 GB
A100 (40 GB version)	~40 GB	about 36 GB

The rest of the GPU memory is the spare room for the chunks of text it is summarizing. Colab's GPU choices change over time; check **Runtime** → **Change runtime type** to see yours.

A step-by-step search

1. Go to huggingface.co/models.
2. In the search box, type `Instruct GGUF`, or a family name plus `GGUF` (for example `Mistral GGUF`).
3. Sort by **Most downloads** or **Trending**. Popular models are well tested.
4. Open a result whose name includes a size (`...B`), `Instruct` and `GGUF`.
5. Click **Files and versions**. Find the file ending in `Q4_K_M.gguF` and check its size against the table above.
6. Skip any model whose page asks you to **log in and agree to terms** before showing the files. It needs an account, and there are plenty of alternatives that don't.
7. Skim the front page of the model for its **licence**. Most allow research use; avoid any that say "non-commercial" if that matters for your work.

Choosing a sorter (embeddings model) works the same way. Search for `embed GGUF`. Sorter files are small, usually under 1.5 GB, so size is rarely a problem. If you read papers in more than one language, pick one described as **multilingual**.

6. How to replace models

The pipeline always refers to its two models by fixed names: `local-analyst` (the writer) and `local-embed` (the sorter). Replacing a model means putting a different file behind one of those names. Nothing else in the project needs changing.

1. **Choose the model** on Hugging Face (section 5). Note its address, shown at the top of its page, for example `bartowski/Mistral-Small-24B-Instruct-2501-GGUF`.
2. **Stop the cloud agent** if it's running.
3. **Open** `replaceModel1.ipynb` **in Colab** and fill in section 2a:
 - `ROLE` : `"local-analyst"` for the writer, or `"local-embed"` for the sorter;
 - `REPO_ID` : the address from step 1;
 - `FILENAME` : leave it blank, and the `Q4_K_M` file is picked for you.
4. **Choose Runtime** → **Run all**. The notebook first checks the model: that the address exists, that it doesn't need a login, and that the file isn't too big for the GPU. Only then does it download the

model and swap it in on your Drive. The old model is kept as, for example, `local-analyst.previous.gguf`.

5. **Start the cloud agent as usual.** It uses the new model automatically. Within a few minutes you should see `[Summary]` ... lines appearing.

6. **Keep it or go back:**

- **Not happy?** Set `ACTION = "rollback"` and run again. You're back on the previous model.
- **Happy?** Set `ACTION = "delete_previous"` and run again, to free the Drive space.

Until you delete the previous model, your Drive holds both (about 30 GB for two writers). To skip keeping the old one, set `KEEP_PREVIOUS = False`, but then there's nothing to roll back to.

Nothing the agent has produced is lost when a model changes: summaries, themes and the research focus are plain text, and the new model carries on from them. `models/MODEL_SOURCES.md` on your Drive records which model is behind each name, and when it changed.

7. Words you'll meet

Word	Plain meaning
GGUF	The file format this project uses. Look for it in model names
GPU	The chip that runs AI models fast
GPU memory / VRAM	The GPU's own memory; the model must fit in it
Instruct / Chat / it	A model trained to follow instructions for what you need
LLM	Large language model: the "writer"
Embeddings model	The "sorter", using mathematical similarity
Q4_K_M (and similar)	How much a model file has been compressed ('Q' is for 'quantizing')
RAM	A computer's normal working memory
24B, 70B ...	Model size in billions of internal settings
SSD	Stands for 'Solid state drive, which is permanent storage